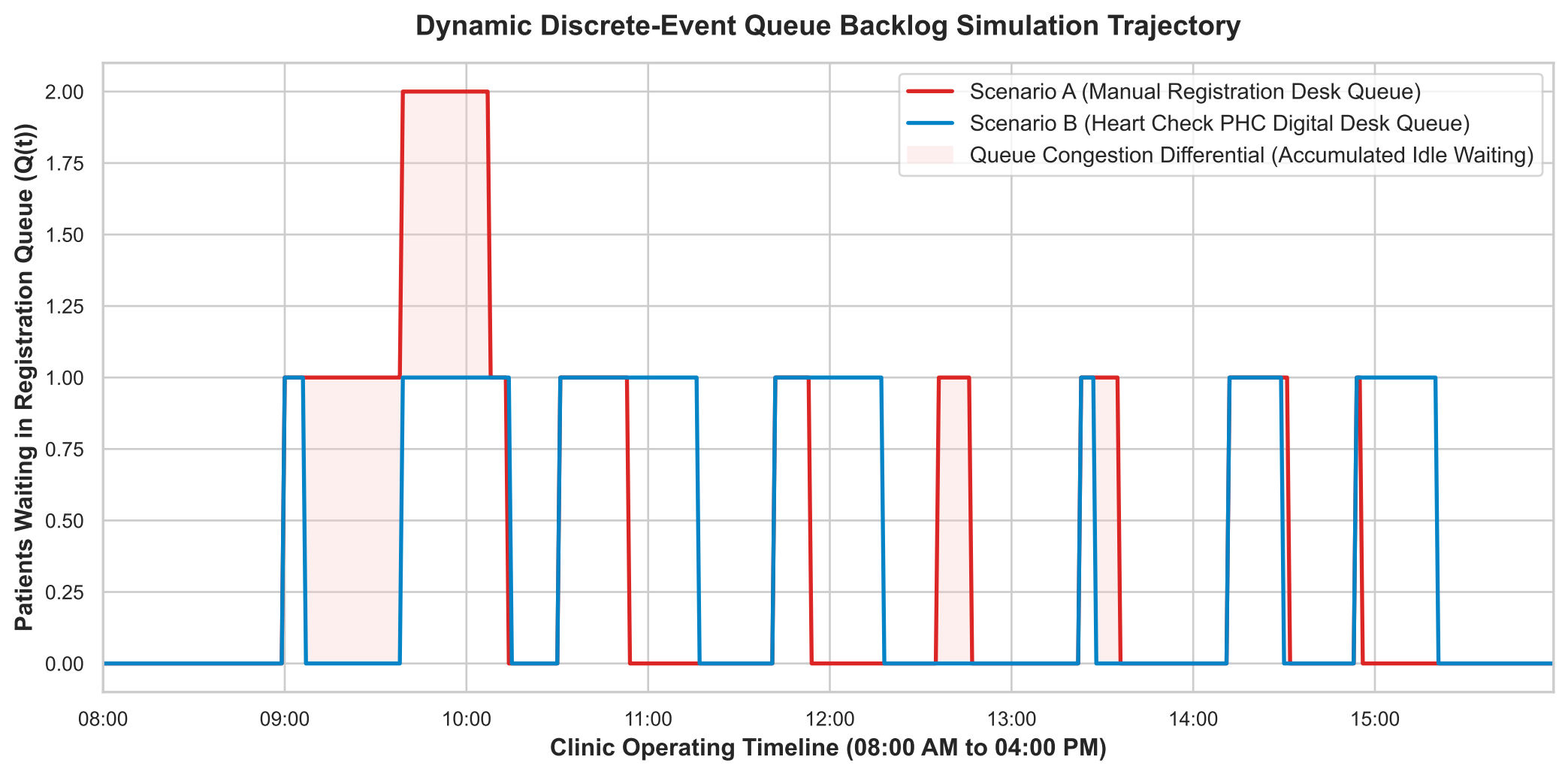
Dynamic Discrete-Event Queue Backlog Simulation Trajectory
Scenario A (Manual Registration Desk Queue)
Scenario B (Heart Check PHC Digital Desk Queue)
Queue Congestion Differential (Accumulated Idle Waiting)
Patients Waiting in Registration Queue (Q(t))
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
08:00
09:00
10:00
11:00
12:00
13:00
14:00
15:00
Clinic Operating Timeline (08:00 AM to 04:00 PM)